

Self-similarity and Scale Invariance

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Self-similarity and Fractals

Two geometric objects are *similar* if they have the same shape regardless of their size, position and orientation.

DEFINITION: A **similarity** $\mathcal{S}$ is a (linear) transformation obtained as a composition of translation, rotation (and/or reflection) and scale transformation.

EXAMPLE: The linear operation $\mathcal{S} = \mathcal{S}(\lambda, \theta, a)$ defined as

$$\mathcal{S} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \lambda \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \quad (1)$$

performs a similarity transformation on elements $x \in \mathbb{R}^2$.

Self-similarity and Fractals

Consider a generalized transformation $\hat{\mathcal{S}}$ defined as the union of n similarity transformations, i.e.

$$\hat{\mathcal{S}}(O) = \mathcal{S}_1(O) \cup \mathcal{S}_2(O) \cup \dots \cup \mathcal{S}_n(O)$$

DEFINITION: A geometrical object O^* is **self-similar** if it exists a (generalized) similarity transformation $\hat{\mathcal{S}}$ such that $\hat{\mathcal{S}}(O^*) = O^*$.

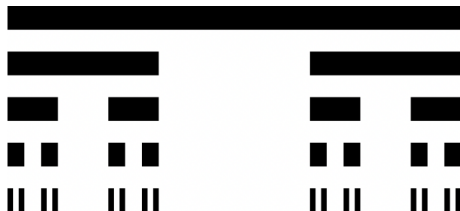
In words, a self-similar object is such that its parts, when magnified by some suitable *scale factor* λ look (exactly) like the whole.

Self-similarity and Fractals

EXAMPLE: Consider $\forall x \in I = [0, 1] \subset \mathbb{R}$ the transformations

$$\mathcal{S}_1(x) = \frac{1}{3}x, \quad \mathcal{S}_2(x) = \frac{1}{3}x + \frac{2}{3}. \quad (2)$$

What is the set $\hat{\mathcal{S}}(I)$ with $\hat{\mathcal{S}} = \mathcal{S}_1 \cup \mathcal{S}_2$? Repeat it to compute $\hat{\mathcal{S}}^n(I)$.



After taking the limit $n \rightarrow \infty$, we obtain the *Cantor set* $\hat{\mathcal{S}}^\infty$, which is a non-trivial self-similar geometric object, an example of a **fractal**.

Fractal Dimension

Common intuitions about the dimension of a geometrical object:

- **Embedding dimension** $d \geq 0$: the spatial dimension of the Euclidean space, such as $\mathbb{R}^d$, in which the object is embedded.
- **Topological dimension** $d_T \geq 0$: if locally, around every point, the object looks like a piece of $\mathbb{R}^{d_T}$ (formally, we need to define Lebesgue's coverings).

REMARK: In general, $d_T \leq d$.

EXAMPLE: The surface of a sphere defined by the equation $x_1^2 + x_2^2 + x_3^2 = 1$ has embedding dimension $d = 3$, but its *topological dimension* is $d_T = 2$.

Fractal Dimension

DEFINITION: The **fractal (or Hausdorff) dimension** d_F is defined as follows:

- 1 cover the object O with d -balls of radius r , i.e.
 $\mathcal{B}_{x_0}(r) = \{x \in \mathbb{R}^d : \|x - x_0\| \leq r\}$;
- 2 define $N(r)$ to be the minimum number of such balls necessary to cover the object O (min. w.r.t. x_0 at fixed r).
- 3 repeat the operation for different r and compute

$$d_F = - \lim_{r \rightarrow 0} \frac{\log N(r)}{\log r}. \quad (3)$$

DEFINITION: A **fractal** is a subset of a Euclidean space whose *fractal dimension* d_F strictly exceeds its *topological dimension* d_T (it can assume non-integer values).

Fractal Dimension

EXAMPLE: Consider the set $S = \{(x_1, x_2) \in \mathbb{R}^2 : x_1(x_2 - x_1^3) = 0\}$.

By covering the curve we have $N(r)r \sim \text{const.}$, which implies $d_F = 1$.

EXAMPLE: Consider the Cantor set: embedded in $d = 1$ but $d_T = 0$ (point-like objects), so we expect $0 \leq d_F \leq 1$.

At each step n of the recursive transformation leading to the Cantor set, we use balls of diameter $2r = 3^{-n}$ and we need $N(r) = 2^n$ balls.

$$d_F = \lim_{n \rightarrow \infty} -\frac{\log N(r)}{\log r} \quad (4)$$

$$= \lim_{n \rightarrow \infty} \frac{-n \log 2}{-n \log 3 - \log 2} \quad (5)$$

$$= \frac{\log 2}{\log 3} \approx 0.63. \quad (6)$$

The Cantor set is a fractal.

Fractal Dimension

EXERCISE: Find transformations to generate the Sierpinski carpet and Koch's curve. Show that their fractal dimensions are respectively $d_F = \frac{\log 8}{\log 3} \approx 1.89 < 2$ and $d_F = \frac{\log 4}{\log 3} \approx 1.26 > 1$.

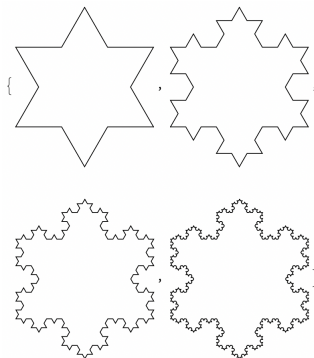
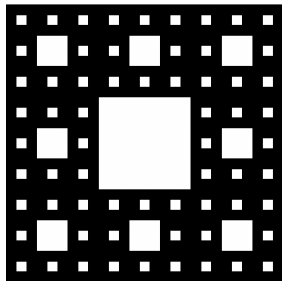


Figure: Sierpinski carpet (left) and Koch snowflake (right).

Scale Invariance and Power Laws

The transformation employed is composed of two steps:

- ① coarse-graining
- ② rescaling



and corresponds to the equation

$$N(r) = 2^{-n} N(r3^{-n}). \quad (7)$$

A general scaling form representing a **self-similar function** is

$$f(x) = \lambda^{-\alpha} f(x/\lambda). \quad (8)$$

with *scale factor* λ (for the Cantor set, $\lambda = 3$ and $\alpha = d_F = \log 2 / \log 3$).

Scale Invariance and Power Laws

DEFINITION: When the self-similarity property holds for every possible scale factor λ , we call it **scale invariance**.

A (diff.) function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *scale invariant* iff there is (diff.) function ϕ such that

$$\frac{f(y)}{f(x)} = \phi(y/x), \quad \forall x, y \in \mathbb{R} \quad (9)$$

Set $y = (1 + \epsilon)x$, for $\epsilon \ll 1$ at first order we get

$$x \frac{f'(x)}{f(x)} = \phi'(1) := -\alpha \quad \implies \quad f(x) \propto x^{-\alpha}, \quad (10)$$

Scale-invariance implies power law functions ($\nRightarrow$).

Scale Invariance and the Renormalization Group

Suppose $\ell = 3^n \rightarrow \pi(\ell)$ and set $2^{-n} = \ell^{-d_F} \rightarrow Z(\ell)$, and assume

$$N(r) = Z(\ell) N(r/\pi(\ell)). \quad (11)$$

Promoting ℓ to be continuous and deriving,

$$\ell \frac{dN(r)}{d\ell} = 0 = \ell \frac{d}{d\ell} (Z(\ell) N(r/\pi(\ell)))$$

we get the **Renormalization Group Equation**

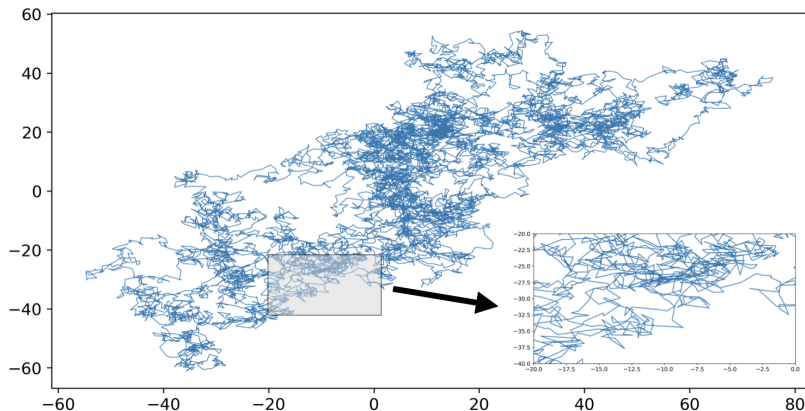
$$\left[\ell \frac{\partial}{\partial \ell} + \beta(\pi) \frac{\partial}{\partial \pi} + \gamma(\ell) \right] N = 0 \quad (12)$$

with $\beta(\pi) = \ell \frac{d\pi}{d\ell}$ and $\gamma(\ell) = \ell \frac{d \log Z(\ell)}{d\ell}$. It tells how a measurable quantity renormalizes when parameters do so under rescaling.

Here $\beta(\pi) = \pi$, and $\gamma(\ell) = -d_F$, so $\pi \partial_\pi N - d_F N = 0 \implies N \sim r^{-d_F}$.

Statistical Scale Invariance

Trajectory of Brownian motion



It's a *random fractal*: a geometric object, realization of a stochastic process, which resembles itself under rescaling.

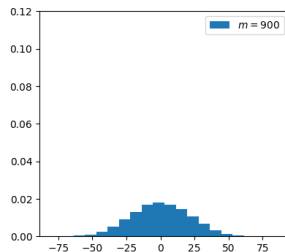
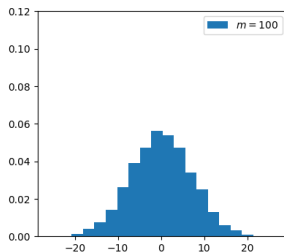
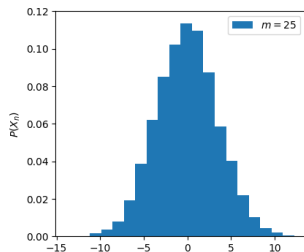
Statistical Scale Invariance

Random walk in $\mathbb{R}^d$: rotational invariant displacement distribution $p(\vec{x})$

The net displacement after m steps, $\vec{X}_m = \sum_{i=1}^m \vec{x}_i$, is a random variable:

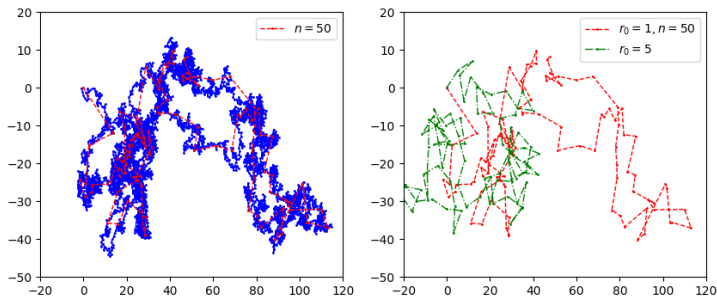
$$\langle \vec{X}_m \rangle_m = \left\langle \sum_{i=1}^m \vec{x}_i \right\rangle_m = \int \prod_{i=1}^m d^d x_i p(\vec{x}_i) \sum_{i=1}^m \vec{x}_i = 0, \quad (13)$$

$$\langle |\vec{X}_m|^2 \rangle_m = \sum_{i=1}^m \sum_{j=1}^m \langle \vec{x}_i \cdot \vec{x}_j \rangle_m = \sum_{i=1}^m \langle |\vec{x}_i|^2 \rangle_m = mr_0^2. \quad (14)$$



Statistical Scale Invariance

A random walk of m steps of size r_0 “looks like” a random walk of m/n steps of (average) size $r_1 = \sqrt{nr_0}$.



Since $N(r) \sim m$ balls for $r \sim r_0$, and $N(r) \sim m/n$ balls for $r \sim r_1$, then

$$N(r_0) = nN(\sqrt{nr_0}) \implies r_0^{-d_F} = nr_0^{-d_F} n^{-d_F/2} \implies d_F = 2. \quad (15)$$

Statistical Scale Invariance

Statistical self-similarity: when a geometric object is sampled from a distribution which is itself self-similar.

EXAMPLE: Consider a Gaussian distribution of displacements,

$$p(\vec{x}) = \frac{1}{(2\pi\sigma^2)^{d/2}} e^{-\frac{|\vec{x}|^2}{2\sigma^2}}. \quad (16)$$

Performing *coarse-graining* and *rescaling* $\implies$ statistical self-similarity

The probability distrib. of displacement $\vec{X}_n = \sum_{i=1}^n \vec{x}_i$ after n steps is

$$p_n(\vec{X}_n) = \int \prod_{i=1}^n \left(d^d x_i p(\vec{x}_i) \right) \delta^{(d)} \left(\vec{X}_n - \sum_{i=1}^n \vec{x}_i \right) \quad (17)$$

$$= \int \prod_{i=1}^n (d\vec{x}_i p(\vec{x}_i)) \int \frac{d^d k}{(2\pi)^d} e^{i\vec{k} \cdot (\vec{X}_n - \sum_{i=1}^n \vec{x}_i)} \quad (18)$$

Statistical Scale Invariance

The real-space integrals are now independent

$$p_n(\vec{X}_n) = \int \frac{d^d k}{(2\pi)^d} e^{i\vec{k} \cdot \vec{X}_n} \prod_{i=1}^n \int d\vec{x}_i p(\vec{x}_i) e^{-i\vec{k} \cdot \vec{x}_i} \quad (19)$$

$$= \int \frac{d^d k}{(2\pi)^d} e^{i\vec{k} \cdot \vec{X}_n} \left[\int d\vec{x} p(\vec{x}) e^{-i\vec{k} \cdot \vec{x}} \right]^n \quad (20)$$

$$= \int \frac{d^d k}{(2\pi)^d} e^{i\vec{k} \cdot \vec{X}_n} \left[e^{-\frac{1}{2}|\vec{k}|^2 \sigma^2} \right]^n. \quad (21)$$

where we used the multivariate version of the gaussian integral identity

$$\int_{-\infty}^{+\infty} dx e^{-\frac{1}{2}ax^2 + bx} = \sqrt{\frac{2\pi}{a}} e^{\frac{1}{2a}b^2}. \quad (22)$$

Performing another Gaussian integral (inverse Fourier transform)

$$p_n(\vec{X}_n) = \int \frac{d^d k}{(2\pi)^d} e^{i\vec{k} \cdot \vec{X}_n} \left[e^{-\frac{1}{2}|\vec{k}|^2 \sigma^2} \right]^n \quad (23)$$

$$= \int \frac{d^d k}{(2\pi)^d} e^{i\vec{k} \cdot \vec{X}_n - \frac{n}{2}|\vec{k}|^2 \sigma^2} \quad (24)$$

$$= \frac{1}{(2\pi n\sigma^2)^{d/2}} e^{-\frac{|\vec{X}_n|^2}{2n\sigma^2}}. \quad (25)$$

It's a coarse-grained RW process with $\sigma_n = \sqrt{n}\sigma$.

Statistical Scale Invariance

Rescaling $\vec{x}' = \vec{X}_n / \sqrt{n}$,

$$\begin{aligned} p(\vec{x}') d\vec{x}' &= p_n(\vec{X}_n) d\vec{X}_n = \frac{1}{(2\pi n\sigma^2)^{d/2}} e^{-\frac{1}{2n\sigma^2} |\vec{X}_n|^2} n^{d/2} d\vec{x}' \\ &= \frac{1}{(2\pi\sigma^2)^{d/2}} e^{-\frac{1}{2\sigma^2} |\vec{x}'|^2} d\vec{x}' \\ &= p(\vec{x}) d\vec{x}. \end{aligned}$$

The distribution $p(\vec{x})$ is invariant under coarse-graining and rescaling.

Since this is true $\forall n$, the RW process is **statistically scale invariant**.

Statistical Scale Invariance

Consider a RW with drift

$$p(\vec{x}) = \frac{1}{(2\pi\sigma^2)^{d/2}} e^{-\frac{1}{2\sigma^2}(\vec{x}-\vec{x}_0)^2}.$$

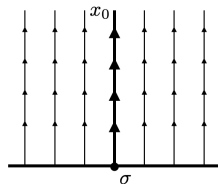
After n steps,

$$p_n(\vec{X}_n) = \frac{1}{(2\pi n\sigma^2)^{d/2}} e^{-\frac{1}{2n\sigma^2}|\vec{X}_n - n\vec{x}_0|^2}.$$

By defining $\vec{x}' = \vec{X}_n/\sqrt{n}$ and imposing $p_n(\vec{X}_n) d\vec{X}_n = p(\vec{x}) d\vec{x}$, the distribution is invariant only if $\vec{x}'_0 = \sqrt{n}\vec{x}_0$.

Different **universality class** compared to RW:

$$\sigma^{(R)} = \sigma, \quad x_0^{(R)} = \sqrt{n}x_0$$

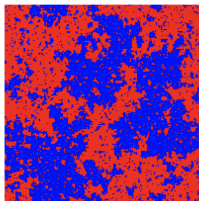


Statistical Scale Invariance

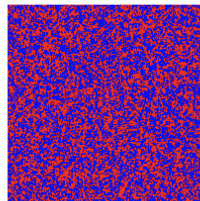
Spin configurations in a ferromagnetic system



$T < T_c$



$T \simeq T_c$



$T > T_c$

Scaling behavior of the **correlation function**:

$$C(r, \xi) \propto r^{-\alpha} Y(r/\xi) \propto \begin{cases} e^{-r/\xi} & \xi < +\infty, \quad r \gg \xi \\ r^{-\alpha} & \xi = \infty, \quad \alpha = d - 2 \end{cases} \quad (26)$$

When $r \rightarrow r' = r/b$, $\Rightarrow C(r, \xi) \propto b^{-\alpha} C(r/b, \xi/b)$

Self-similarity at T_c ($\xi \rightarrow +\infty$): $C(r) \propto b^{-\alpha} C(r/b)$.

Dimensional Analysis and Scaling

A physical phenomenon is described by an equation

$$x_0 = f(x_1, \dots, x_k) \quad (27)$$

with $k + 1$ governing parameters $x_1, \dots, x_k$ and x_0 .

It can be found a *complete, dimensionally independent* subset $\{x_1, \dots, x_m\}$ of them ($m \leq k$), such that

$$[x_{m+i}] = \left[\prod_{j=1}^m x_j^{r_{m+i,j}} \right] = \prod_{j=1}^m [x_j]^{r_{m+i,j}}, \quad i = 1, \dots, k - m \quad (28)$$

where $\{r_{i,j}\}$ are rational numbers.

Dimensionless versions of x_0 and $\{x_{m+1}, \dots, x_k\}$:

$$\pi_0 = \frac{x_0}{\prod_{j=1}^m x_j^{r_{0,j}}}, \quad \pi_i = \frac{x_{m+i}}{\prod_{j=1}^m x_j^{r_{m+i,j}}} \quad (29)$$

Dimensional Analysis and Scaling

In terms of dimensionless governing parameters,

$$\pi_0 = f(x_1, x_2, \dots, x_m; \pi_1, \dots, \pi_{k-m}). \quad (30)$$

The l.h.s. is dimensionless, so $\{x_1, \dots, x_m\}$ must disappear, i.e.

$$\pi_0 = \tilde{f}(\pi_1, \dots, \pi_{k-m}). \quad (31)$$

THEOREM (Buckingham):

When a complete relationship between dimensional physical quantities is expressed in dimensionless form, the number of independent quantities that appear in it is reduced from the original k to $k - m$, where m is the maximum number of the original k governing parameters that are dimensionally independent.

Dimensional Analysis and Scaling

EXAMPLE: Motion of a simple pendulum described by the equation of motion $\theta = f(t, \theta_0, m, \ell, g)$.

Using Buckingham's theorem:

- $k = 5$ independent quantities $\{t, \theta_0, m, \ell, g\}$,
- $m = 3$ represent a complete base $\{t, \ell, m\}$ of independent dimensions (i.e. time T , length L and mass M).

$\Rightarrow k - m = 2$ dimensionless quantities: θ_0 and $\tau = t\sqrt{g/\ell}$.

The equation of motion can be expressed as

$$\theta = \tilde{f}(\theta_0, \tau) \quad [= \theta_0 \sin \tau] \quad (32)$$

REMARK: The mass m has correctly disappeared from the equation!

Dimensional Analysis and Scaling

EXAMPLE: A ball is thrown vertically and reaches the ground after time t^* . Known quantities: the initial height h , the initial velocity v_0 , the mass m of the ball and the gravity acceleration g . Compute the equation of motion.

Using Buckingham's theorem:

- $k + 1 = 5$ governing parameters $\{t, h, v_0, m, g\}$,
- $m + 1 = 3$ dimensionally independent $\{t, h, m\}$,

$\implies k - m = 2$ dimensionless quantities: $\pi_0 = t\sqrt{g/h}$ and $\pi_1 = v_0/\sqrt{gh}$

One can write

$$t^* = \sqrt{\frac{h}{g}} \tilde{f} \left(\frac{v_0^2}{gh}, h, m \right). \quad (33)$$

where $\tilde{f}$ is now dimensionless.

Dimensional Analysis and Scaling

Since $\tilde{f}$ should not depend on h, m , we obtain the physical law

$$t^* = \sqrt{\frac{h}{g}} \phi\left(\frac{v_0}{\sqrt{gh}}\right). \quad (34)$$

REMARK: We found a “*scaling law*” because under rescaling $h \rightarrow h' = h/b$

$$t^*(h, v_0, g) = b^{1/2} t^*(h/b, v_0 b^{-1/2}, g). \quad (35)$$

Asymptotic behavior:

- for $v_0 \rightarrow 0$, t^* is finite, then

$$\phi(x) \sim \text{const.} \quad x \rightarrow 0$$

- for $v_0 \rightarrow \infty$, t^* is independent of h , then

$$\phi(x) \sim x, \quad x \rightarrow +\infty$$

Dimensional Analysis and Scaling

Let's check the exact form of $\phi(x)$: the equation of motion is

$$y(t) = h + v_0 t - \frac{1}{2}gt^2, \quad (36)$$

and setting $y(t^*) = 0$, we find

$$t^* = \frac{v_0}{g} + \sqrt{\left(\frac{v_0}{g}\right)^2 + \frac{2h}{g}} \quad (37)$$

$$= \sqrt{\frac{h}{g}} \left(\frac{v_0}{\sqrt{gh}} + \sqrt{\frac{v_0^2}{gh} + 2} \right), \quad (38)$$

that is $\phi(x) = x + \sqrt{2 + x^2}$ (in agreement with the asymptotic analysis).

Intermediate Asymptotics

Consider the behavior of $\Phi(\pi_1, \dots, \pi_{k-m})$ when $\pi_1 \rightarrow 0$, three cases:

- **complete similarity**: Φ tends to a non-zero finite limit $\Phi(0, \pi_2, \dots, \pi_{k-m})$;
- **incomplete similarity**: a non-zero limit does not exist but Φ has a power-law behavior

$$\Phi = \pi_1^{\eta_1} \tilde{\Phi} \left(\frac{\pi_2}{\pi_1^{\eta_2}}, \dots, \frac{\pi_{k-m}}{\pi_1^{\eta_{k-m}}} \right),$$

with $\tilde{\Phi}$ regular and η are *anomalous dimensions*;

- Φ has **no finite limit** and no power-law behavior.

REMARK: In the previous example, $\Phi(\pi_1)$ goes to a constant as $\pi_1 \rightarrow 0$.